# Semantic Change Guided Remote Sensing Image Generation

Nurşize Turan, Orkhan Karimli, Melisa Tanrıkulu, and Alperen Tolga Karaçam

***Abstract*—Generating remote sensing images that reflect specific, controlled semantic changes remains a significant challenge due to the scarcity of annotated datasets and the difficulty of preserving spatial context. In this work, we propose a semantic change-guided image generation framework based on Latent Diffusion Models (LDMs). We introduce a dual-conditioning mechanism that utilizes a frozen DINOv2 encoder and a Perceiver Resampler to adapt pre-change visual features into the diffusion process via cross-attention. To effectively eliminate background drift in unchanged regions, we implement an Inpainting Hard Constraint during inference. Furthermore, we critically analyze the limitations of standard evaluation metrics in resource-constrained scenarios. We demonstrate how small test set sizes introduce significant bias into Fréchet Inception Distance (FID) scores, and show that extreme class imbalance in satellite imagery penalizes segmentation-based metrics (mIoU), even for ground truth data.**

***Index Terms*—Remote Sensing Image Generation, Latent Diffusion Models (LDMs), Semantic Change Detection, Dual-Conditioning Mechanism, Perceiver Resampler.**

## I. INTRODUCTION

REMOTE sensing imagery is crucial for applications ranging from urban planning and environmental monitoring to disaster response. Analyzing temporal changes in these images is essential for understanding dynamic surface processes. However, developing robust change detection algorithms requires massive annotated datasets, which are scarce and labor-intensive to create. This has spurred interest in synthetic data generation to augment training sets and improve model generalization.

Generative models have achieved remarkable success in computer vision. While Generative Adversarial Networks (GANs) [1] were traditionally the standard for image synthesis, they often suffer from training instability and limited diversity. The advent of Denoising Diffusion Probabilistic Models (DDPMs) [2] and Latent Diffusion Models (LDMs) [3] has revolutionized the field, offering superior image quality and controllability. In remote sensing, these models have been applied to tasks including super-resolution, in-painting, and text-to-image generation [4].

Despite these advancements, generating remote sensing images that reflect specific, controlled changes remains a significant challenge. Most text-to-image models lack the precise spatial control required to simulate localized changes while preserving the unchanged context. Even with semantic maps, ensuring that generated changes are realistic and blend seamlessly with the environment is difficult. Existing methods often struggle to maintain consistency between the pre-change context and the post-change generation, leading to artifacts.

In this work, we explore the application of LDMs for semantic change-guided remote sensing image generation. We generate post-change images conditioned on a pre-change image and a semantic change map. Drawing inspiration from advancements like CC-Diff++ [5], we incorporate a Perceiver-based Resampler [6] to adapt features extracted by a frozen DINOv2 encoder [7]. We present our experimental attempts and discuss the challenges of adapting these architectures, particularly under computational constraints. Our results indicate that while the proposed architecture has theoretical merit, achieving high-fidelity generation requires careful tuning and substantial resources.

The main contributions of this paper are:

- We investigate a framework for semantic change-guided generation based on Latent Diffusion Models.
- We implement a dual-conditioning mechanism utilizing a DINOv2 encoder and a Perceiver Resampler.
- We analyze the limitations and challenges encountered, offering insights for future research in resource-constrained environments.

The remainder of this paper is organized as follows. Section II reviews related work in remote sensing image generation and diffusion models. Section III details the proposed methodology, including the architecture and conditioning mechanisms. Section IV presents the experimental setup. Section V presents the results, and a discussion of the limitations. Section VI gives brief explanation of implementations. Finally, Section VII concludes the paper.

## II. RELATED WORKS

### A. Generative Paradigms in Remote Sensing

The field of remote sensing image generation has witnessed a paradigm shift from Generative Adversarial Networks (GANs) [1] to Diffusion Models. Early works primarily utilized GAN variants, such as CycleGAN [8] and Pix2Pix [9], for tasks like image-to-image translation, super-resolution, and domain adaptation. While effective for specific translation tasks, GANs often struggle with training instability and mode collapse, limiting their ability to capture the diverse and complex textures inherent in satellite imagery. Recently, Denoising Diffusion Probabilistic Models (DDPMs) [2] and Latent Diffusion Models (LDMs) [3] have demonstrated superior capability in modeling complex data distributions. By learning to reverse a gradual noising process, these models generate high-fidelity images with greater diversity. In remote sensing, LDMs have been successfully adapted for various applications, leveraging their stable training dynamics to synthesize realistic terrain, vegetation, and urban structures that were previously challenging for adversarial approaches.

### B. Controllable Synthesis: From Text to Layout

Unconditional generation, while impressive, lacks the controllability required for practical remote sensing applications. Consequently, research has focused on conditioning diffusion models on auxiliary inputs.

*1) Text-Driven Generation:* Initial efforts, such as RSDiff [10] and Text2Earth [4], introduced text-to-image synthesis for remote sensing. Text2Earth, for instance, utilized a global-scale dataset to align textual descriptions with satellite imagery. However, text prompts often suffer from spatial ambiguity; a prompt like “industrial area near a river” does not specify the exact location or orientation of the features, limiting its utility for precise simulation.

*2) Spatially-Guided Generation:* To address the limitations of text, recent works have incorporated spatial constraints. CRS-Diff [11] conditions the generation process on semantic segmentation maps, ensuring that the synthesized image strictly adheres to a predefined cartographic layout. Similarly, CC-Diff++ [5] employs a two-stage approach where a Large Language Model (LLM) first generates a spatial layout (bounding boxes) from a text prompt, which then guides the diffusion model. These methods demonstrate that injecting spatial information—often via cross-attention or control adapters—is essential for generating geographically accurate and semantically coherent scenes.

### C. Temporal and Reference-Guided Generation

While layout-to-image models can generate realistic static scenes, simulating temporal changes (e.g., urbanization, deforestation) presents a unique challenge: the model must generate new content that is not only realistic but also consistent with a *pre-existing* reference image. This requires a “reference-guided” generation approach, where the model must preserve the unchanged background context while seamlessly integrating changes defined by a semantic map. ChangeBridge [12] represents a significant step in this direction, enabling multi-conditional generation where a post-change image is synthesized based on a pre-change image and a condition (text, map, or difference mask). Our work builds upon this concept but specifically investigates a robust dual-conditioning mechanism. Unlike general layout-to-image methods that generate from scratch, we focus on the feature preservation aspect, utilizing a frozen DINOv2 encoder [7] and a Perceiver Resampler [6] to explicitly model and adapt the visual context of the pre-change image for the diffusion process.

## III. Methodology

We propose a semantic change-guided remote sensing image generation model based on LDMs [3]. Our goal is to generate a realistic post-change image given a pre-change image and a semantic change map. The architecture leverages a pre-trained frozen encoder for image feature extraction, a Perceiver-based resampler for feature adaptation, and a conditional U-Net for the diffusion process. Our approach is illustrated in Fig. 1.

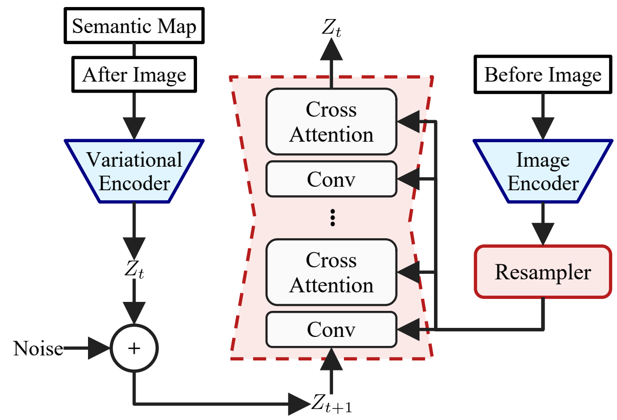


Fig. 1. Overview of the proposed architecture.

### A. Pre-change Image Encoding

To capture the semantic context of the pre-change image, we utilize the DINOv2 ViT-L/14 model [7] as a frozen image encoder. DINOv2 is a self-supervised vision transformer that learns robust visual features. Given a pre-change image $I_{pre}$, we extract the patch tokens and the class token from the last layer. These features provide a rich representation of the scene's structure and content, crucial for maintaining consistency in the generated post-change image.

### B. Resampler

Directly using the variable-length sequence of features from the ViT encoder can be computationally expensive and may not align well with the U-Net's cross-attention layers. To address this, we employ a Resampler module inspired by Flamingo [13] and Perceiver [6]. The Resampler takes the features from the DINOv2 encoder and a set of learnable latent queries as input. Through a cross-attention mechanism, it aggregates the visual information into a fixed number of output tokens. This allows the model to focus on the most relevant visual cues while maintaining a constant computational cost.

### C. Semantic Conditioning and Generation

The core generation process is handled by a U-Net architecture operating in the latent space of a pre-trained Variational Autoencoder (VAE). The semantic change map $M_{sem}$, indicating regions of change, is resized to match the spatial dimensions of the VAE latents. It is then concatenated channel-wise with the noisy latents $z_t$ at each timestep $t$. This spatial conditioning ensures that generated changes align precisely with the provided semantic layout.

Simultaneously, the output tokens from the Resampler are injected into the U-Net via cross-attention layers. This global conditioning guides the generation process to respect the visual style and content of the pre-change image in unchanged regions, while plausibly synthesizing new content in changed regions.

### D. Training Objective

The model is trained using the standard diffusion objective. We predict the noise $\epsilon$ added to the latent representation of the ground truth post-change image $I_{post}$. The loss function is defined as:

$$\mathcal{L} = \mathbb{E}_{z_0,t,c_{pre},c_{sem},\epsilon\sim\mathcal{N}(0,1)} \left[\|\epsilon - \epsilon_\theta(z_t, t, c_{pre}, c_{sem})\|_2^2\right] \tag{1}$$

where $z_t$ is the noisy latent at timestep $t$, $c_{pre}$ represents the condition from the pre-change image (via Resampler), $c_{sem}$ is the semantic change map, and $\epsilon_\theta$ is the U-Net denoiser.

## IV. Experiments

### A. Dataset and Preprocessing

We evaluate our method on the SECOND (Semantic Change detection Dataset) [14], a benchmark dataset designed for semantic change detection in aerial imagery. The dataset consists of 4,662 pairs of high-resolution ($512 \times 512$) images collected from diverse platforms and sensors over cities such as Hangzhou, Chengdu, and Shanghai. Each sample includes a pre-change image, a post-change image, and a pixel-level semantic change map. The annotations cover six primary land-cover classes: non-vegetated ground surface, tree, low vegetation, water, buildings, and playgrounds. These classes form 30 distinct change categories, including non-change, reflecting real-world distributions of land-cover transitions.

For our experiments, we randomly split the dataset into training, validation, and test sets with a ratio of 70%, 20%, and 10%, respectively. To ensure reproducibility, a fixed random seed was used for the split. During preprocessing, all images are resized to the model's input resolution. We normalize the pixel values to the range $[-1, 1]$ using a mean and standard deviation of 0.5 for all channels. The semantic change maps are processed to align with the spatial dimensions of the latent features.

### B. Ablation Studies

To determine the optimal architecture for preserving spatiotemporal consistency, we conducted an ablation study focusing on the input conditioning mechanism of the pre-event satellite imagery. We compared two distinct strategies for injecting the pre-event context into the diffusion process:

*1) Experimental Setups:*

- **Method A: Channel Concatenation (Early Fusion):** In this configuration, the pre-event image is concatenated channel-wise with the post-event image (noise) and the semantic map before being passed into the Variational Autoencoder (VAE). The combined latent representation is then fed into the U-Net. This approach relies on the model learning to disentangle the conditions solely through the initial convolutional layers.
- **Method B: Latent Feature Injection (Cross-Attention):** In this proposed configuration, the pre-event image is processed independently through a frozen DinoV2 encoder to extract high-level semantic features. These features are then projected via a Perceiver Resampler and injected into the U-Net via Cross-Attention layers. This allows the model to leverage robust, pre-trained representations of the pre-event structure.

*2) Ablation Results:* Table I presents the quantitative comparison between the two conditioning strategies. While the baseline Channel Concatenation (Method A) provides a simpler architecture, our results demonstrate that the DinoV2-based Cross-Attention mechanism (Method B) offers superior preservation of fine-grained structural details. Specifically, Method B achieves a significantly lower **LPIPS** score (0.5051 vs. 0.5625) and a higher **mIoU** (17.96% vs. 13.26%), indicating that the generated images are perceptually closer to the ground truth and adhere more strictly to the input segmentation masks.

We observe that Method A achieves a lower (better) **FID** score (201.81 vs. 261.93). We attribute this to the trade-off between structural constraint and unconstrained texture synthesis: the baseline model, being less conditioned on the spatial layout, is prone to hallucinating realistic-looking but semantically incorrect textures that statistically resemble the training distribution. In contrast, our method prioritizes semantic fidelity, resulting in outputs that are spatially accurate relative to the input conditions, despite the penalty in the distributional metric.

TABLE I
QUANTITATIVE COMPARISON BETWEEN CHANNEL CONCATENATION AND OUR DINOV2 APPROACH.

| Method | LPIPS ↓ | FID ↓ | mIoU ↑ |
|---|---|---|---|
| Channel Concatenation (VAE) | 0.5625 | **201.8108** | 0.1326 |
| DinoV2 + Cross-Attention (Ours) | **0.5051** | 261.93 | **0.1796** |

## V. Results

### A. Quantitative Analysis

We monitored the training progression of our proposed model (Method B) over 50 epochs. Table II demonstrates the stability of the training process. The model achieves its best trade-off between perceptual quality (FID) and structural similarity (SSIM) around the final epochs, though fluctuations in FID suggest further tuning may be required.

TABLE II
EVALUATION RESULTS AT DIFFERENT TRAINING EPOCHS.

| Epochs | PSNR (dB) | SSIM | LPIPS | FID | IoU |
|---|---|---|---|---|---|
| 10 | 11.98 | 0.1264 | 0.5714 | 341.60 | 0.6450 |
| 20 | 13.63 | 0.1635 | 0.5428 | 279.43 | 0.5836 |
| 30 | 13.57 | 0.1656 | 0.5273 | 237.82 | 0.5878 |
| 40 | 13.07 | 0.1584 | 0.5199 | 236.02 | 0.6008 |
| 50 | 13.89 | 0.1764 | 0.5265 | 293.37 | 0.5899 |

FID typically requires at least around 2000 samples (standard practice) to be statistically significant. Due to the small size of our test set ($\approx 300$), standard FID scores are subject to significant bias and instability. While we attempted to mitigate this by using the larger training set as the reference distribution to approximate *realism*, the metric ultimately remains unreliable for this sample size. Therefore, we prioritize qualitative results (visual inspection).

- FID enhancement after using train set and validation set as reference set is **FID : 212.33**

We also benchmarked our final model against the current state-of-the-art method, ChangeBridge [12] as shown in Table III. It is important to note that the mIoU scores for both our method and the Ground Truth are notably low in our evaluation pipeline (21.26% for Real Images), indicating that the segmentation "judge" model penalizes the extreme class imbalance present in the dataset (e.g., small objects vs. large background areas). Relative to the Real Data Upper Bound, our model achieves approximately 84.5% performance relatively.

TABLE III
FINAL QUANTITATIVE COMPARISON.

| Method | PSNR ↑ | SSIM ↑ | FID ↓ | mIoU ↑ |
|---|---|---|---|---|
| *Baselines (Reported)* | | | | |
| ChangeBridge [Zhao et al.] | - | - | 98.42 | 73.46% |
| *Our Evaluation Pipeline* | | | | |
| Real Images (Ground Truth) | - | 1.000 | 0.00 | 21.26% |
| **Ours (Proposed)** | **14.30** | **0.188** | **261.93** | **17.96%** |

The notable discrepancy between the reported baseline mIoU and our measured Ground Truth mIoU (21.26%) indicates also that the SegFormer network used for evaluation was not trained sufficiently to robustly handle the dataset's severe class imbalance, resulting in a suppressed upper bound for all models. Consequently, our method's performance is best understood in relative terms, where it successfully recovers 84.5% of the semantic layout achievable by the Real Images within this constrained evaluation pipeline. We achieve this competitive relative performance after only 50 training epochs, whereas the ChangeBridge [12] baseline requires 3000 epochs to reach convergence. This reduction in computational cost highlights our model's limitations, demonstrating that it can capture essential semantic structures, even if the absolute segmentation metrics are limited by the "judge" model's capacity.

## *B. Qualitative Analysis*

Figure 2 visual result confirms that the model synthesizes post-event imagery that adheres to the layout defined by the semantic map mostly coherent with the environment.

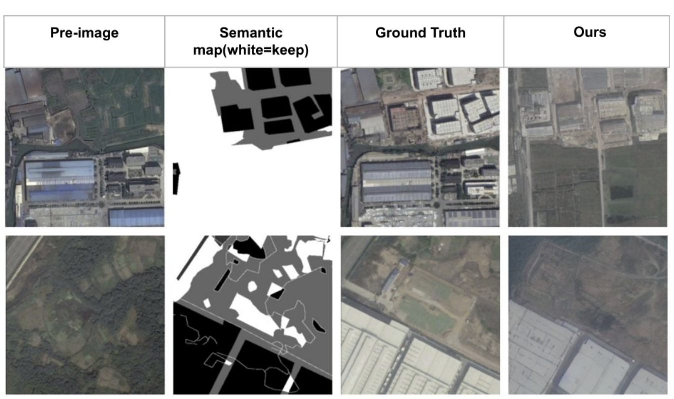


Fig. 2. Generated image samples from the input of the pre-image and semantic map (semantic map provided as grayscale mask)

The generated images generally did not preserve the texture of the pre-event image in unchanged regions as marked in Figure 3 with the red squares.

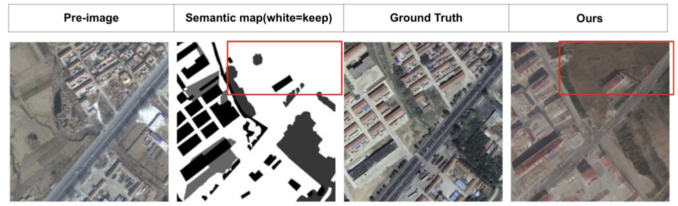


Fig. 3. Samples of generated images that did not keep the white regions unchanged (semantic map provided as grayscale mask)

We analyzed the impact of the "Inpainting Hard Constraint" on the preservation of non-changed regions to handle this problem. Initial experiments utilizing a standard masked cross-attention approach yielded suboptimal results, where unchanged regions drifted significantly from the ground truth. To address this, we introduced a hard constraint mechanism (Inpainting) during the inference stage to force consistency in non-white semantic regions. Visual impact of the Inpainting Hard Constraint on performance can be seen in Figure 4 below.

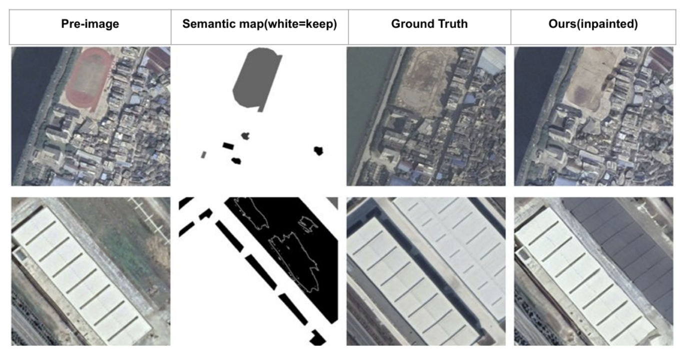


Fig. 4. Generated image samples from the input of the pre-image and semantic map using inpainting constraint during the inference stage (semantic map provided as grayscale mask)

However even after that, limitations remain in generating rare classes; for example, specific textures for water or dense vegetation occasionally vanish or appear blurred due to their scarcity in the training data as some problematic examples can be seen in the Figure 5.

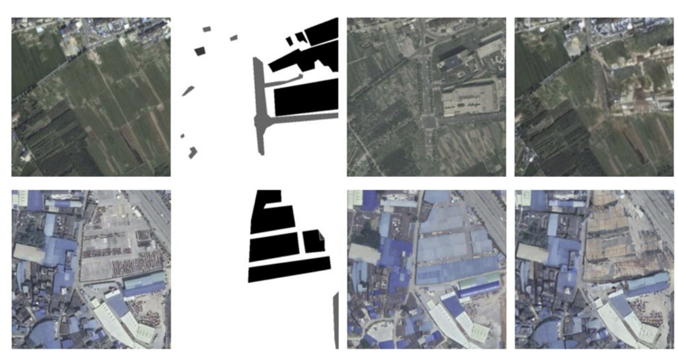

Fig. 5. Generated samples are still problematic, must-change regions contain blurry structures and it is not possible to determine which classes the objects belong to (semantic map provided as grayscale mask)

## VI. Implementations

We employed a U-Net base model using the diffusers library, which provided the LDM and VAE implementations and weights. However, the resampler, training loop, data loading, and visualization pipelines were implemented from scratch using PyTorch and various utility packages. For evaluation, we used standard packages for LPIPS, CLIP, and SSIM (via scikit-image). We manually implemented the FID metric using a torchvision backbone and scipy, as well as the mIoU metric and SemanticSegmentationDataset using native PyTorch operations. Visualization was handled by a custom pipeline built with PyTorch and Matplotlib

## VII. Conclusion

In this study, we presented a Conditional Spatiotemporal Diffusion Bridge for remote sensing image generation. By integrating a custom 7-channel input adaptation with a DinoV2-based Perceiver Resampler, we aimed to improve the semantic consistency of generated satellite imagery.

Our results demonstrate that:

- **Input Conditioning Matters:** The use of a dedicated visual encoder (DinoV2) and Cross-Attention (as explored in the ablation study) is critical for preserving the high-frequency details of the pre-event image compared to simple channel concatenation.
- **Structural consistency is robust,** though the model struggles with extreme class imbalance, leading to the loss of rare objects in the generated output.
- **Metric sensitivity and limitation:** Standard segmentation metrics (mIoU) may underrepresent generation quality in unbalanced datasets, as evidenced by the low scores on real ground truth data. Also, FID scores were unstable due to the small size of our test set ($\approx 300$ samples), necessitating the use of the training set as a reference distribution to approximate realism.

Future work will focus on addressing class imbalance through weighted loss functions and refining the inpainting constraints to further eliminate background drift in unchanged regions.

## References


[1] I. J. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, and Y. Bengio, "Generative adversarial nets," *Advances in neural information processing systems*, vol. 27, 2014.

[2] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," *Advances in neural information processing systems*, vol. 33, pp. 6840–6851, 2020.

[3] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, "High-resolution image synthesis with latent diffusion models," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 10 684–10 695.

[4] C. Liu, K. Chen, R. Zhao, Z. Zou, and Z. Shi, "Text2earth: Unlocking text-driven remote sensing image generation with a global-scale dataset and a foundation model," *IEEE Geoscience and Remote Sensing Magazine*, 2025.

[5] M. Zhang, Y. Liu, Y. Liu, Y. Zhao, and Q. Ye, "Cc-diff++: Spatially controllable text-to-image synthesis for remote sensing with enhanced contextual coherence," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 63, pp. 1–16, 2025.

[6] A. Jaegle, F. Gimeno, A. Brock, O. Vinyals, A. Zisserman, and J. Carreira, "Perceiver: General perception with iterative attention," in *International conference on machine learning*. PMLR, 2021, pp. 4651–4664.

[7] M. Oquab, T. Darcet, T. Moutakanni, H. Vo, M. Szafraniec, V. Khalidov, P. Fernandez, D. Haziza, F. Massa, A. El-Nouby *et al.*, "Dinov2: Learning robust visual features without supervision," *arXiv preprint arXiv:2304.07193*, 2023.

[8] J.-Y. Zhu, T. Park, P. Isola, and A. A. Efros, "Unpaired image-to-image translation using cycle-consistent adversarial networks," in *Proceedings of the IEEE international conference on computer vision*, 2017, pp. 2223–2232.

[9] P. Isola, J.-Y. Zhu, T. Zhou, and A. A. Efros, "Image-to-image translation with conditional adversarial networks," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2017, pp. 1125–1134.

[10] A. Sebaq and M. ElHelw, "Rsdiff: Remote sensing image generation from text using diffusion model," *Neural Computing and Applications*, vol. 36, no. 36, pp. 23 103–23 111, 2024.

[11] D. Tang, X. Cao, X. Hou, Z. Jiang, J. Liu, and D. Meng, "Crs-diff: Controllable remote sensing image generation with diffusion model," *IEEE Transactions on Geoscience and Remote Sensing*, 2024.

[12] Z. Zhao, C. Wu, D. Wang, H. Chen, and Z. Zheng, "Changebridge: Spatiotemporal image generation with multimodal controls for remote sensing," *arXiv preprint arXiv:2507.04678*, 2025.

[13] J.-B. Alayrac, J. Donahue, P. Luc, A. Miech, I. Barr, Y. Hasson, K. Lenc, A. Mensch, K. Millican, M. Reynolds *et al.*, "Flamingo: a visual language model for few-shot learning," *Advances in neural information processing systems*, vol. 35, pp. 23 716–23 736, 2022.

[14] K. Yang, G.-S. Xia, Z. Liu, B. Du, W. Yang, M. Pelillo, and L. Zhang, "Semantic change detection with asymmetric siamese networks," *arXiv preprint arXiv:2010.05687*, 2020.